# Analyzing & Storytelling

## From raw data to clean, human readable data ready for visualization

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# Where we are

**01**

Mar 25

**Sensing & Collecting**

*Complete*

**02**

Apr 1

**Analyzing & Storytelling**

***Today***

**03**

Apr 8

**Visualizing & Presenting**

Today: clean your data, find the story angles, sketch out visualization concepts.

Between sessions: build at least 3 visualizations. Session 3 is wrap up and presentations.

# Today - Session 2

**9:30** **Data check-in**

What did you collect? What worked, what didn't? Review datasets together briefly.

**9:50** **Data cleaning in Excel**

Handling errors, nulls, and outliers. Aggregating by time interval if needed. Building a summary sheet.

**10:20** **Demo: AI-assisted image analysis**

How the AI Assisted Analysis Tool extracts structured data from an image sequence. A second data stream alongside sensor readings.

**10:45** **Short break**

**10:55** **Narrative arc frameworks**

Three frameworks for finding the story in a dataset. What makes data worth communicating.

**11:15** **Activity**

Find 2-3 story angles in your own data. Sketch out visualization concepts and explore Flourish and AI driven JS.

**12:00** **Wrap**

Share your story angles. Expectations and tools for Session 3.

# Cleaning your data

### Look at it first

Scroll through the whole dataset before touching anything. Note the range, any obvious gaps, any values that look wrong. Build intuition before building formulas.

### Handle errors, nulls, and outliers

Missing readings are normal - sensors occasionally fail. Label them as missing, don't fill with zeros. A temperature spike to 96°C or a motion reading 20 seconds after you stopped the script. Real events or sensor errors? Document your reasoning.

### Aggregate to intervals (if needed)

10-second readings are noisy. Average by minute, or count events per 5-minute window. The interval depends on your phenomenon - faster events need finer resolution.

### Build a summary sheet

Add a second tab with summary statistics: mean, min, max, count. This is what you'll visualize from. Keep the raw data intact on the first tab.

# AI-assisted image analysis

A second data stream from any timestamped image sequence - phone photos, RPi camera output, or footage from a walk. Same workflow for text.

**1 Image sequence** → **2 Run n iterations** → **3 Modal consensus** → **4 Excel output**

Any folder of timestamped images. The prompt defines what to measure: people count, activity level, spatial quality, object presence, etc.

Each image is analyzed multiple times by a local language model. Multiple runs per image account for model errors.

The most common response across runs is returned, with a confidence score. Low-confidence items are flagged for manual review.

Structured numeric data per image, timestamped. Merges directly with sensor data by timestamp.

***Key design principle: the prompt is the instrument specification. What you ask the model to measure is the measurement decision.***

Runs entirely locally. No data is sent externally.

github.com/henrylevesque/AI_Assisted_Analysis_Tool · DOI: 10.5281/zenodo.14932653

# Finding the story in your data

## The comparison

What is different between two conditions, places, or times? The story is in the gap: this vs. that, then vs. now, here vs. there.

*e.g. Morning vs. evening temperature. Motion on weekdays vs. weekends. Sound levels near a door vs. far from it.*

## The pattern

What repeats? What's predictable? The story is the rhythm - or the disruption.

*e.g. Temperature drops every time the HVAC cycles off. Motion clusters at 10 minutes before and after the hour. Sound spikes on a schedule.*

## The surprise

What did you not expect to find? What contradicts your assumption? The story is the gap between prediction and observation or data and expectation.

*e.g. The 'quiet' corner of the studio is actually the most active. The outdoor temperature is more stable than the indoor temperature.*

# Visualization tools

Choose the tool that fits your story. You should have at least three visualizations by Session 3.

## Start here

**Excel / Google Sheets and Flourish (flourish.studio)**

Time series charts, scatter plots, summary tables. Fast, easy, standard, representations of data. Annotatable narrative charts, line stories, bar races. Exports as shareable link - portfolio-embeddable.

## Go further

**p5.js (p5js.org)**

Data-driven generative sketches. Visuals that change based on your data. iframe-embeddable in portfolio pages.

**Rawgraphs (rawgraphs.io)**

Good for exploratory chart types: alluvial, bump, matrix. No code required.

## Optional

**D3.js or Figma**

D3.js for full control over custom data visualization. Significant learning curve - start here only if you have prior JS experience. Figma for layout and infographic design, combining charts with editorial text and visual framing.

# Between now and April 8

1. **Refine your story angle**

   Of the 2–3 angles you found today, pick the one that's most interesting. You can include the others as supporting context.

2. **Build at least 3 visualizations**

   They should support your story. Mix types - time series, distribution, comparison. Start in Flourish or Excel, push at least one into p5.js or another tool, even if it means abstracting the concept or data.

3. **Develop a short narrative**

   What does the data show? Why does it matter? What would you do differently if you ran the study again? 2–3 sentences, not a wall of text.

4. **No fixed slide format for the final**

   Present in whatever way best communicates your story. Visuals, annotated screenshots, or a live walkthrough. Session 3's final hour is presentations and open discussion.

# Questions?

*Let's look at your data.*